

1 - false : eg

$$\exists u, v \in \mathbb{R}^3, u = \{1, 2, 3\}, v = \{1, 1, \sqrt{2}\},$$
$$\|u\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \quad \|v\| = \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = \sqrt{4}$$

$$u - v = \{1, 2, 3\} - \{1, 1, \sqrt{2}\}$$

$$= \{0, 1, -\sqrt{2}\} \rightarrow \|u - v\| = \sqrt{0^2 + 1^2 + (-\sqrt{2})^2}$$
$$= \sqrt{1 + 2} = \underline{\underline{3}} \neq 0$$

2 - yes, as both vectors have to be in the same vector space to be orthogonal.

3 - If it were in $\mathbb{R}^2$, yes, as there would only be one direction as to how the vectors could be perpendicular. However, in 3-D space 2 vectors have 2 different ways it could be perpendicular.

$$4 - \begin{aligned} u^T v &= 0 \\ u^T w &= 0 \\ &u^T (2v - 5w) \\ &= (u^T 2v) - (u^T 5w) \\ &= 2(u^T v) - 5(u^T w) \\ &= 2 \cdot 0 - 5 \cdot 0 \rightarrow \text{yes} \end{aligned}$$

5 -

$$u^T(v+w) = 0$$

$$u^T v + u^T w = 0$$

$$u^T(v-w) = 0$$

$$u^T v - u^T w = 0$$

$$u^T v + u^T w = 0$$

$$u^T v - u^T w = 0$$

$$\hline$$

$$2u^T v = 0$$

$$u^T v = 0$$

6 - $uv = 0$
 $uw = 0$

$$av + bw$$

$$\hookrightarrow u(av + bw)$$

$$= (u(av) + u(bw))$$

$$= (a(uv) + b(uw)) = a \cdot 0 + b \cdot 0 = 0$$

7 -

$$uv = 0$$

$$uw = 3$$

$$u(v + cw) = (uv + u(cw)) = 0$$

$$= 0 + c(wu) = 0$$

$$= 0 + c \cdot 3 = 0$$

$$3c = 0$$

$$c = 0$$

$$\underline{\underline{c = 0}}$$

8 -

$$uv = 0$$

$$wu = (3v - 2u)u$$

$$w = 3v - 2u$$

$$= 3vu - 2u^2$$

$$= -2u^2 \rightarrow \text{no, dependent on } \vec{u}$$